Introduction

The advancement of medical imaging technology has paved the way for enhanced diagnostic tools in neuro-oncology, particularly in the detection and classification of brain tumors. This project explores an integrated approach to brain tumor segmentation and classification using MATLAB, leveraging both classical image processing techniques and modern machine learning algorithms. The primary objective is to accurately delineate tumor regions from brain MRI scans and subsequently classify different tumor types based on distinctive image features.

To achieve segmentation, the project implements a suite of image processing techniques including thresholding and morphological operations. These methods work in tandem to effectively isolate the tumor regions from surrounding tissues, thereby simplifying the complexity inherent in brain images. To effectively train the machine learning models employed in this work, we implement a preprocessing technique based on Histogram of Oriented Gradients (HOG) features. HOG features are well-regarded for capturing the essential structural characteristics and edge directions of the tumor, which are critical for accurate classification. For the classification phase, three machine learning algorithms, such as, Support Vector Machines (SVM), Random Forests (RF), and Neural Networks (NN) are employed. These algorithms are trained on a labeled dataset, enabling them to learn and recognize subtle differences among various brain tumor types. By combining the precise segmentation techniques with the powerful discriminative capabilities of these classifiers, the project demonstrates a comprehensive framework for aiding clinical diagnosis, potentially leading to more timely and accurate treatment decisions.

In summary, this project not only highlights the versatility and efficiency of MATLAB in handling complex medical imaging tasks but also underscores the potential of integrated image processing and machine learning methodologies in advancing the field of brain tumor diagnostics.

State of the art

Segmentation Techniques and Feature Extraction

Traditionally, techniques such as Otsu thresholding, region growing, watershed segmentation, and morphological operations have provided a basis for tumor region delineation in MRI images. These methods benefit from well-established mathematical formulations and are easily implemented in environments like

MATLAB. However, their sensitivity to intensity variations and image noise has prompted the development of more robust strategies. Recent literature increasingly favors hybrid approaches that augment classical segmentation with advanced feature extraction. For instance, Ullah et al. (2023) [6] introduced a novel method that integrates handcrafted descriptor, such as intensity, texture (including Histogram of Oriented Gradients), and shape feature, with convolutional neural network (CNN) architectures to achieve improved segmentation performance on the BraTS dataset . This hybridization leverages domain-specific insights while also benefiting from the automatic feature learning capabilities of deep networks.

Machine Learning-Based Classification

In parallel with segmentation, classification methods have evolved from traditional statistical approaches to ensemble and deep learning models. Several studies have demonstrated that classifiers such as Support Vector Machines (SVM), Random Forests (RF), and neural networks, when trained on robust feature representations (e.g., HOG features), can achieve exceptional accuracy. A recent study by Kumar and colleagues (2024) [7] evaluated multiple machine learning algorithms for brain tumor classification on MRI datasets and reported that a Random Forest classifier achieved an accuracy as high as 99% when paired with a simple image loading feature extraction method. This work underscores the potential of leveraging handcrafted features like HOG within a machine learning framework to differentiate between various tumor types effectively.

Hybrid and Ensemble Approaches

The state of the art is trending toward multi-phase and hybrid frameworks that combine the strengths of both handcrafted features and deep learning. Zahoor et al. (2022) [8] proposed a two-phase deep hybrid framework that first employs a boosted and ensemble learning scheme to detect tumor regions and then integrates dynamic features from a custom CNN (BRAIN-RENet) with static HOG features for refined tumor classification. Their approach reported detection accuracies exceeding 99.5% and classification accuracies around 99.2%, which illustrate the power of combining complementary feature types.

In another noteworthy contribution, developed a topological optimized network that merges transfer learning with recurrent architectures. By employing advanced hyperparameter tuning through optimization techniques such as Elephant Herding Optimization, their model achieved segmentation and classification metrics in the high 99% range. Such results indicate that integrating state-of-the-art deep learning models with robust handcrafted feature extraction not only mitigates the challenges inherent in conventional methods but also pushes the boundaries of diagnostic accuracy.

Summary

Overall, the latest literature demonstrates a clear shift toward hybrid models in brain tumor analysis. By fusing classical image processing techniques with deep learning and ensemble methods, researchers have achieved significant improvements in segmentation accuracy and classification reliability. These approaches, which often incorporate robust feature extraction methods such as HOG alongside CNN-based architectures, have proven capable of handling the variability in tumor appearance and imaging conditions. This evolution promises more reliable computer-aided diagnosis systems that can ultimately enhance treatment planning and patient outcomes.

Methodologies employed

The following section aims to explain in detail the methodology employed for segmentation and classification and the techniques used.

Segmentation

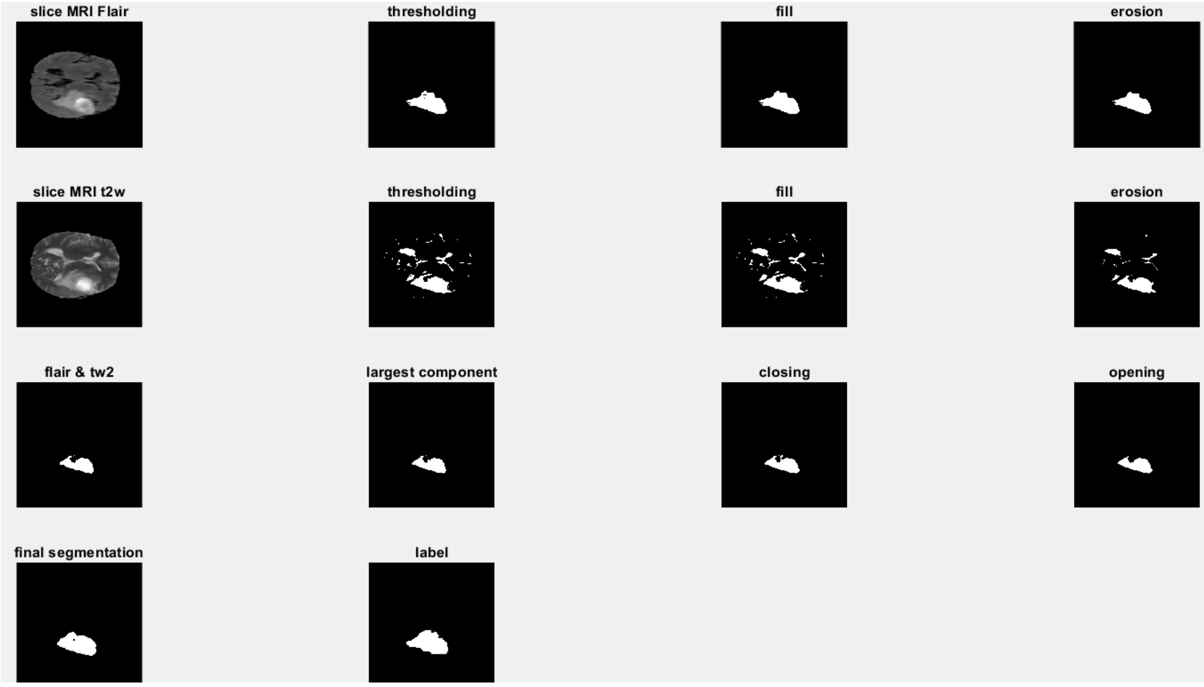
In our proposed approach for brain tumor segmentation, we designed a MATLAB program that integrates several image processing and segmentation techniques to delineate tumor regions from MRI scans using these two papers as a guide [1,2]. We used the BRATS dataset [3] containing NIFTI files of brain tumor MRI scans with labels corresponding to the human made segmentation. Each scan is composed of 155 slices and 4 modalities: Flair, T1w, T1gd, T2w. The method is described in detail as follows:

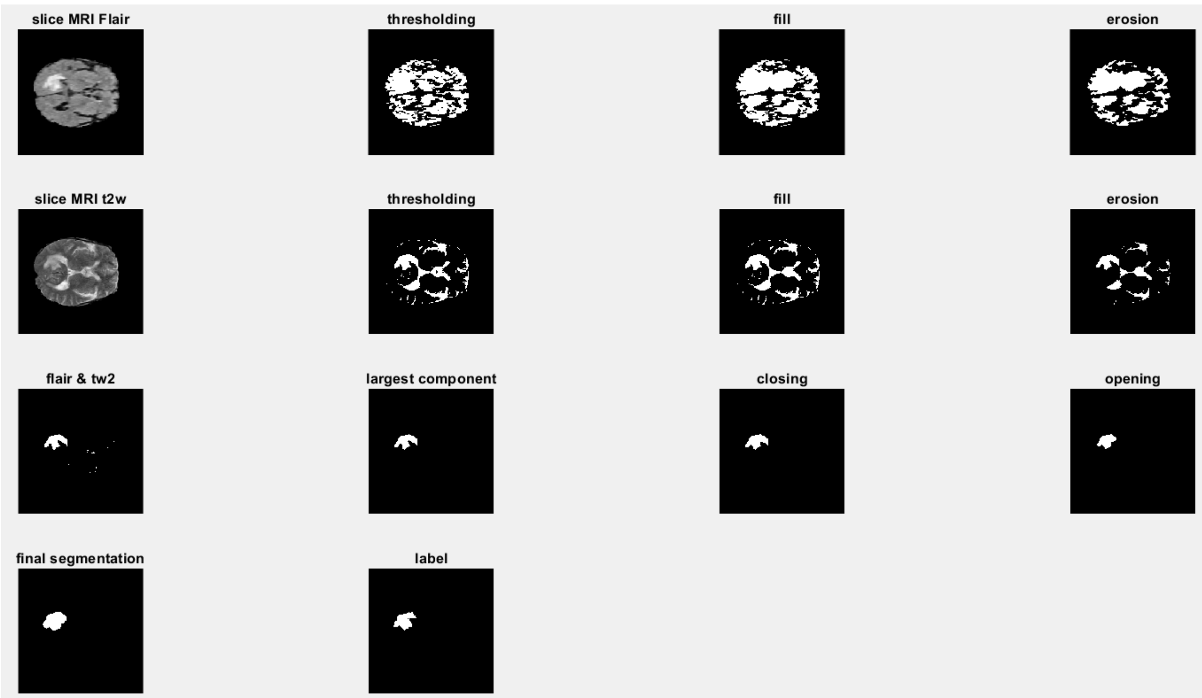
1. For each file we perform the segmentation iterating over each slice.
2. For our purposes we select the Flair and T2w modalities as we think they provide contrasting characteristics that helps our goal.
3. Then we apply thresholding using a global threshold which we selected via trial and error.
4. After that we perform two morphological operations: fill and erosion using a disk as a structural element.
5. We combine the Flair and T2w slice segmentations with a logic AND operator in order to select regions in common.
6. We combine the slices to obtain a 3D segmentation.
7. Then we select the largest connected component from the newly obtained volume.
8. Eventually we apply three morphological operations: closing, opening and dilation using as a structural element a sphere.

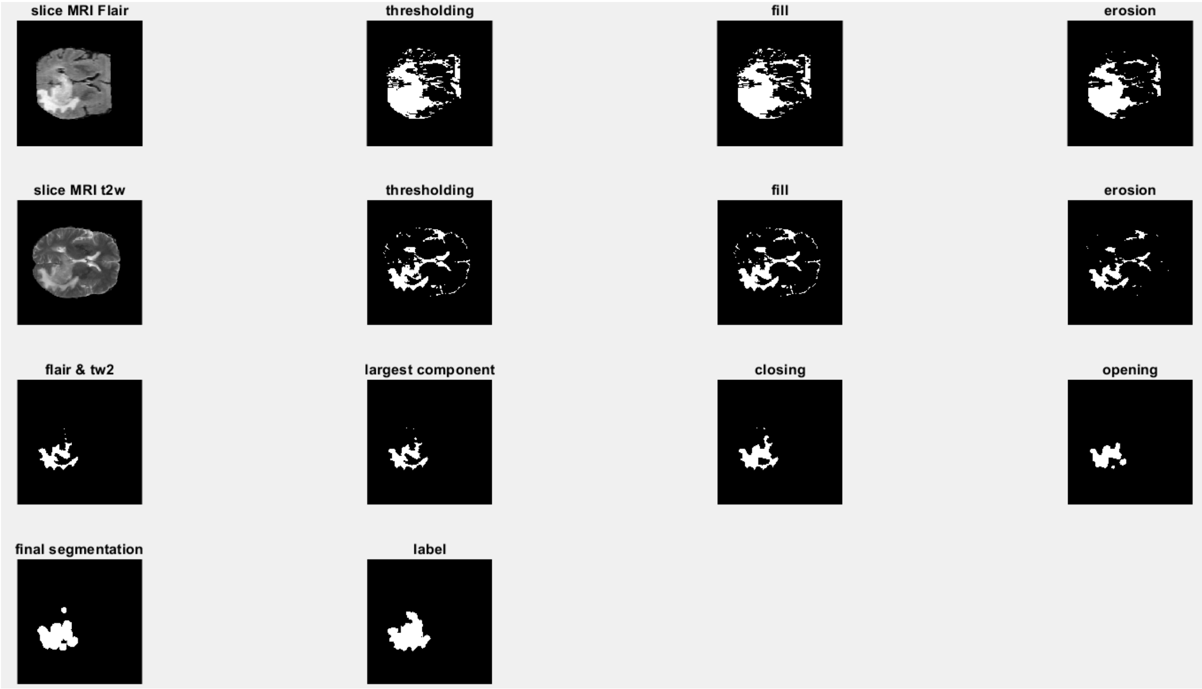
In order to evaluate the quality of the segmentation we use Dice Score as evaluation metric comparing it with the label. From a test pool of 20 scans we averaged a Dice Score of 0.65 with some segmentations reaching scores over 0.8.

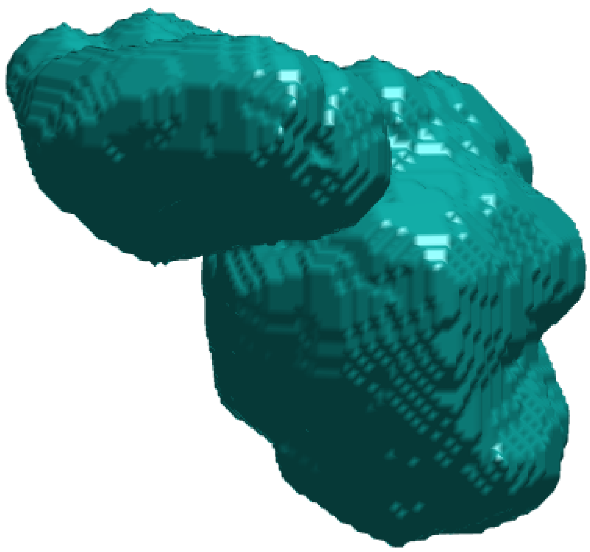
In order to visualize the steps we select a representative slice and we show the segmentation step by step with intermediate results. There is also the possibility to see the 3D segmentation using volumeViewer Matlab function.

The following images represent a demo of the code.






slice MRI Flair
thresholding
fill
erosion
slice MRI t2w
thresholding
fill
erosion
flair & tw2
largest component
closing
opening
final segmentation
label



Classification using machine learning models

Data Preparation and Feature Extraction

The proposed classification framework begins by reading annotated MRI images from designated training and testing folders. The training folder contains 1695 samples of MRI scans in JPG format, while the test folder contains 246 images [4]. Ground truth labels are stored in accompanying JSON files, which are parsed to extract file names and associated labels. Each image is first converted to grayscale and resized to a uniform 64×64 pixel format to standardize the input across the dataset. To capture discriminative image characteristics, Histogram of Oriented Gradients (HOG) features are computed for each image. The extracted feature vectors are concatenated into a training matrix, while the corresponding labels are stored in a cell array and subsequently converted to a categorical type for consistency. An identical process is followed for the test set to ensure that both training and test data are represented by comparable feature spaces.

Classifier Training

The framework employs three distinct classification models to evaluate the effectiveness of the extracted features:

1. **Support Vector Machine (SVM):**
   An SVM classifier is configured using a radial basis function (RBF) kernel with an automatically scaled kernel parameter. The multiclass problem is addressed using an error-correcting output codes (ECOC) framework. This classifier is trained on the HOG feature matrix from the training set, with the goal of learning discriminative boundaries between the different brain tumor classes.
2. **Random Forest:**
   A Random Forest model with 200 trees is implemented. The ensemble learning approach aggregates decisions from multiple decision trees to enhance robustness and generalization. The model is trained on the same feature matrix, with class labels provided as categorical data, to effectively capture complex decision boundaries in the high-dimensional feature space.
3. **Neural Network:**
   A simple feed-forward neural network is constructed. Here, the training feature matrix is transposed (to conform to the neural network's expected input dimensions), and labels are converted to one-hot encoding. A single hidden layer containing 10 neurons is used, and the network is trained using standard backpropagation. Although this network is relatively shallow, it is

sufficient to benchmark the performance against traditional machine learning models given the size and nature of the feature vectors.

In this study, the classifiers are designed to distinguish among four categories: **glioma**, **meningioma**, **pituitary**, and **no tumor**.

- **Glioma:** Malignant tumors originating from glial cells in the brain.
- **Meningioma:** Typically benign tumors that arise from the meninges, the protective layers surrounding the brain and spinal cord.
- **Pituitary Tumor:** Tumors that develop in the pituitary gland, potentially affecting hormone production.
- **No Tumor:** Normal brain images without any abnormal growth.

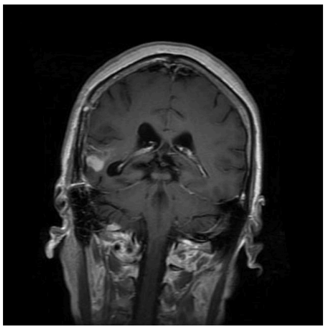
Model Evaluation and Interactive Prediction

After training, the models are evaluated on the test dataset. Predictions from each model are compared against the ground truth labels to compute accuracy metrics. The SVM and Random Forest models directly yield categorical predictions, while the neural network outputs a probability vector that is mapped back to class labels. The corresponding accuracies are calculated and printed.

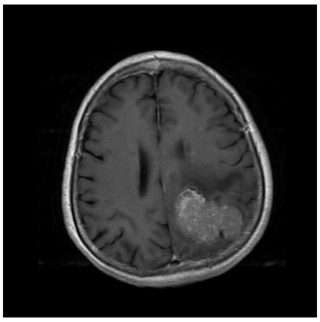
In addition, an interactive component is provided. The user is prompted to select a single image for prediction via a graphical file dialog. The selected image undergoes the same preprocessing and feature extraction steps as the training images. Each trained model then produces a prediction for the input image. For the neural network, the predicted class is determined by finding the maximum output activation (adjusted for indexing differences). Finally, the predictions from all three models are displayed alongside any available ground truth label retrieved from the JSON files.

The following examples show the accuracy achieved by these models as well as a demo prediction of four images selected by the user.

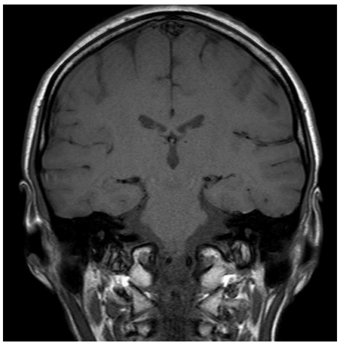
```
SVM Accuracy: 88.00%
Random Forest Accuracy: 88.40%
Neural Network Accuracy: 81.60%
Do you want to select an image for prediction? (y/n):
```



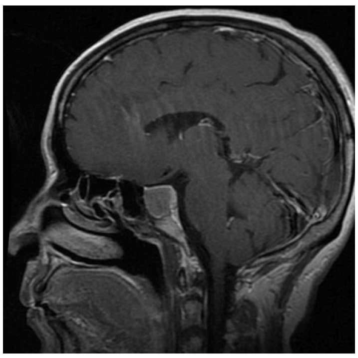
```
Predictions for the selected image:
Ground Truth Label: Glioma
SVM Prediction: Glioma
Random Forest Prediction: Glioma
Neural Network Prediction: Glioma
```



```
Predictions for the selected image:
Ground Truth Label: Meningioma
SVM Prediction: Meningioma
Random Forest Prediction: Meningioma
Neural Network Prediction: Meningioma
```



```
Predictions for the selected image:
Ground Truth Label: No Tumor
SVM Prediction: No Tumor
Random Forest Prediction: No Tumor
Neural Network Prediction: No Tumor
```



```
Predictions for the selected image:
Ground Truth Label: Pituitary
SVM Prediction: Pituitary
Random Forest Prediction: Pituitary
Neural Network Prediction: Pituitary
```

Our implementation is able to achieve accuracies ranging from 80% to 90% for all models. These results are comparable to what the authors of the paper were able to achieve [5].

Conclusion

In summary, this project presents a robust framework for the segmentation and classification of brain tumors using MATLAB. By integrating image processing techniques like thresholding, morphological operations, and machine learning classifiers (SVM, Random Forest, and a neural network based on HOG features), the system effectively distinguishes between glioma, meningioma, pituitary tumors, and normal brain images. The experimental results demonstrate high accuracy and robustness across different classifiers, with ensemble methods such as Random Forest achieving particularly promising performance.

Overall, the framework not only facilitates accurate brain tumor detection and classification but also offers an interactive prediction module for practical clinical evaluation. Future work may extend this approach by incorporating additional imaging modalities, refining feature extraction techniques, and exploring deeper neural network architectures to further enhance diagnostic precision and generalizability.

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Fabio Dell'Aquila
Jacopo Pagano